Take-home submission · Automated post-call workflow

# Every customer call, turned into structured action — automatically.

An end-to-end post-call automation that fires the moment a call ends: it reconciles Gong and Grain, classifies and protects exec calls, extracts insight with Claude, updates Salesforce without ever writing blind, and stores every run for retrospective analysis.

Designed by **Omri Gonen** · GTM Engineer Take-Home · 2026

**n8n** orchestration | **Claude** extraction | **Salesforce** source of truth | **Gong + Grain** capture | **BigQuery** analytics | **Slack** routing

**16**
orchestration nodes, contracted end-to-end

**2 → 1**
sources (Gong + Grain) reconciled into one pipeline

**6**
production-grade edges hardened, not hand-waved

**8**
BigQuery tables answering retro questions in SQL

THE THESIS

## The five problems that actually make this hard

“When a call ends, send the transcript to AI and add it to Salesforce” is the answer a chatbot gives. The judgment lives in the edges. The whole design is built around these five.

01

### Two sources, one call

Gong recordings, Grain-only transcripts, sometimes both — needs a normalize + race-safe dedup layer.

02

### Restricted routing

Internal exec calls must land in an admin-only area. Classification fails **closed** — when unsure, restrict.

03

### Matching, not guessing

From a list of attendee emails to the right account / contact / opportunity — a new face at a known account is auto-created; only net-new accounts need a human gate.

04

### Proving AI quality

A confidence gate that doesn’t trust the model’s self-rating, plus an eval loop that measures drift.

05

### Retro-ready storage

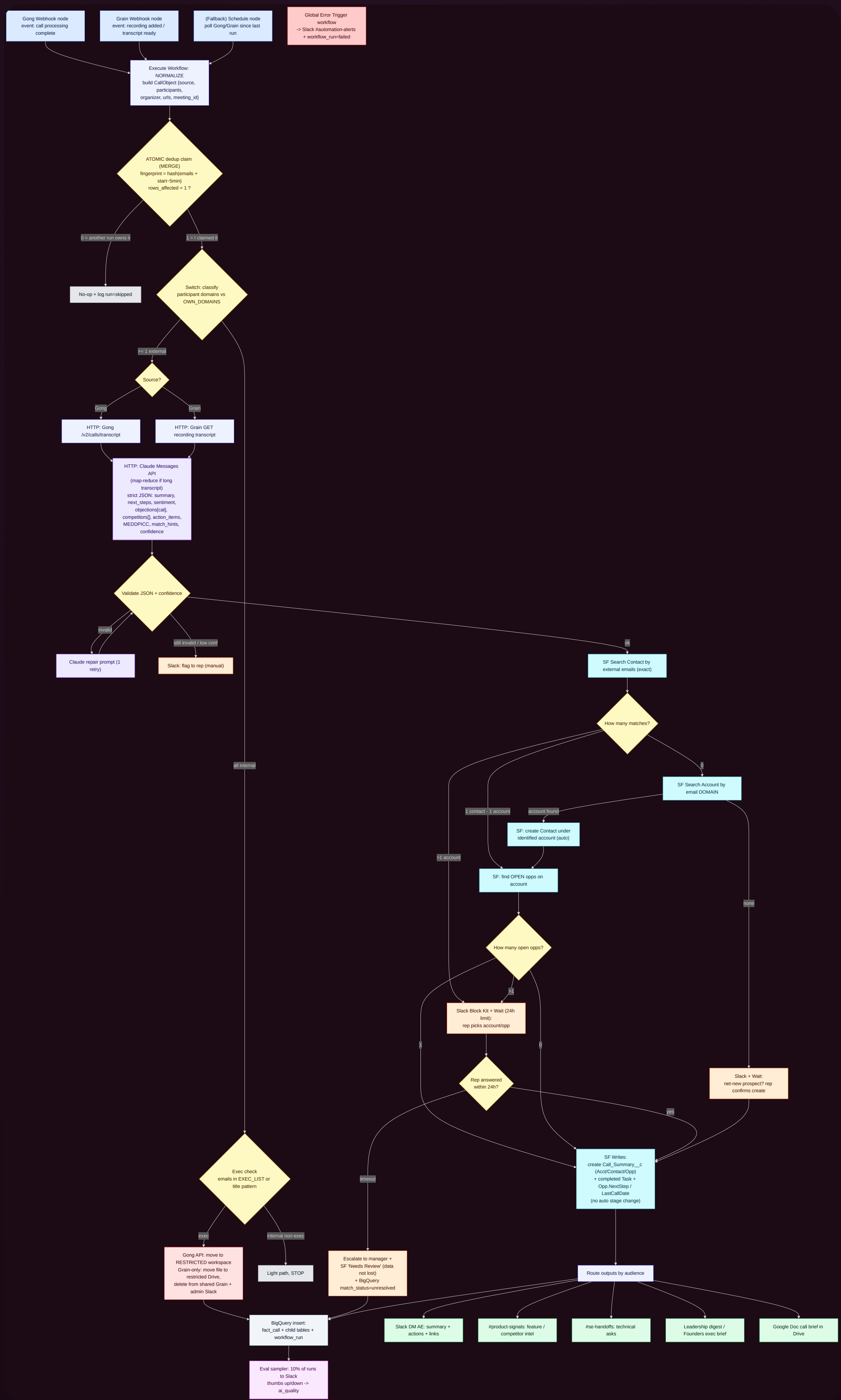
Every run stored structurally so the business can ask “every Q2 pricing-objection call” in SQL.

PART 1 · THE WORKFLOW

## End-to-end flow

Two webhook front doors (Gong + Grain) converge into one normalized pipeline. Built in **n8n** — the only in-scope tool with webhooks, branching, a wait-for-human node, raw HTTP for

Gong/Grain, native Salesforce/Slack/BigQuery nodes, and a global error workflow, all self-hostable for exec-call privacy.



blue: triggers · indigo: process · yellow: decisions · purple: AI · cyan: Salesforce · red: restricted/security · orange: human-in-the-loop · green: outputs · slate: storage

THE BAR

# Node contract — event · data in/out · failure path

For every step: the node, the exact event it reacts to, what moves in and out, and what happens when it goes wrong.

| # | NODE (TYPE · OP) | TRIGGER / EVENT | DATA IN | DATA OUT | FAILURE PATH |
|---|---|---|---|---|---|
| 1 | **Webhook · Gong** | call processing complete | callId, parties[] | raw event | missed → Schedule poll |
| 2 | **Webhook · Grain** | recording added | recording_id, participants[] | raw event | missed → Schedule poll |
| 3 | **Code · Normalize** | — | raw event | CallObject | malformed → error wf |
| 4 | **Code · Fingerprint** | — | CallObject | + fingerprint | — |
| 5 | **BigQuery · MERGE claim** | — | fingerprint | rows_affected | dup → stop (atomic) |
| 6 | **Switch · Classify** | — | participants[] | branch ext/int | low-conf → review |
| 7 | **IF · Exec check** | — | emails vs EXEC_LIST | branch | uncertain → restrict (fail-closed) |
| 8 | **HTTP · Transcript** | — | source_id | transcript text | not ready → Wait+retry → partial |
| 9 | **HTTP · Claude (map/reduce)** | — | transcript | insight JSON | invalid → repair → manual |
| 10 | **IF · Validate** | — | insight JSON | pass/fail | fail → Slack manual |
| 11 | **SF · Search Contact** | — | external emails | contact/account | 0 → domain lookup; >1 → HITL |
| 12 | **SF · Account by domain → Contact** | — | email domain | account + new contact | known acct → auto-create contact; none → confirm |
| 13 | **Switch · Match + Wait** | — | results | 1/0/many | many → Wait+pick; timeout → escalate |
| 14 | **SF · Create Call_Summary__c** | — | insight + IDs | record id | error → error wf |
| 15 | **Slack/Docs · Route** | — | insight + links | messages | — |
| 16 | **BigQuery · Insert** | — | fact + children | row ids | always writes workflow_run |

PRODUCTION-GRADE

# Six edges that separate a demo from a system

Where the design refuses to be naive — each is wired, not hand-waved.

### 01 · Race-safe dedup

Dedup is a **write, not a read**: a BigQuery `MERGE` claims the fingerprint atomically. `rows_affected=1` → I own the call; `0` → another run already has it, stop. The simultaneous-webhook race becomes a deterministic winner.

### 02 · Human gate can't lose calls

The wait-for-rep node has a **24h limit**. On timeout it escalates to the manager **and** persists the insight as `Needs Review` with `match_status=unresolved`. The data survives even if the human doesn't answer.

### 03 · Long-transcript map-reduce

Long calls route through **Split → Claude (map) → Merge → Claude (reduce)**; short calls take the single-shot path. Cost, latency and context limits stay bounded.

### 04 · Trustworthy quality signal

Don't gate on the model grading its own homework. Gate on **structural truth** — required fields filled, competitor names in an allow-list, match-hint emails in the attendee list — plus a 30-call golden set as regression on every prompt change.

### 05 · Exec exposure — named

Grain transcribes **before** classification, so a brief shared-space window exists. The flow scrubs at detection; the real zero-exposure fix is an upstream Grain workspace-default policy. Flagged as a dependency, not hidden.

### 06 · Personal-email prospects

External attendees on free-email domains (gmail/outlook) skip the domain→account match and go straight to contact-email match / human confirm — so they're never mis-bucketed.

PART 2 · DISCOVERY

# Getting the right output to the right team

Not assumed — discovered. For each group: who to talk to, what to ask, the output we co-define, and how we validate it. Outputs start as a Slack message; only what proves used graduates to a Salesforce field or dashboard.

### 🎯 Account Executives

**WHO** 2 AEs (one top performer, one new) + Sales Manager.

**ASK** "What do you retype into SF by hand? What do you forget? What do you need before the next call?"

**OUTPUT** Auto-drafted summary + next steps + pre-filled NextStep + risk flag. Kill CRM admin.

**VALIDATE** Run on 10 of their own past calls — would they have sent it as-is? Track accept-vs-edit rate.

### 🛠 Solution Engineering

**WHO** 1–2 SEs + SE lead.

**ASK** "Which technical / integration / security asks get lost between the AE call and your follow-up?"

**OUTPUT** A #se-handoffs card: technical requirements, integrations named, security asks + transcript link.

**VALIDATE** SEs confirm extracted asks on samples; track handoffs accepted without pinging the AE.

### 📦 Product

**WHO** 1–2 PMs + a product analyst.

**ASK** "How do you hear customer pain today? What taxonomy do you already use?" — adopt theirs.

**OUTPUT** Structured feature-request + competitor-intel feed, tagged to their themes, into #product-signals + BigQuery.

**VALIDATE** PMs label a sample — does AI tagging agree (inter-rater)? Trends match their gut on known topics.

### 📈 Sales Leadership

**WHO** VP / Head of Sales.

**ASK** "What do you want to know across all deals weekly? Which deals need your attention?"

**OUTPUT** Weekly digest — objection mix, competitor trend, at-risk deals, sentiment by stage. Aggregate, not per-call.

**VALIDATE** Does "at-risk" match deals they already worried about? Calibrate thresholds before automating.

### 🚀 Founders

**WHO** The founder(s) directly.

**ASK** “Which accounts/themes do you personally want eyes on? Strategic signal vs noise? One line or a brief?”

**OUTPUT** Short exec brief for strategic / at-risk accounts + emerging market & competitor narrative. Exec calls stay admin-only.

**VALIDATE** Founder reads 3 briefs, says what to cut/add. Success = they forward it or act on it, not just receive it.

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PART 3 · STORAGE & RETRO ANALYSIS

## A structure that answers questions backwards

**BigQuery** as the analytics store; Salesforce stays the operational source of truth. The retro questions are flexible, time-series, trending queries across all calls — a columnar-warehouse job, in-scope (Google Workspace), and it feeds Looker / Sheets / Gemini for the team dashboards.

## Schema (star)

**`fact_call`** FACT
call_id, fingerprint, source, call_date, account_id, opp_id, rep_id, call_type, sentiment, confidence, urls, match_status, workflow_run_id

**`call_objections`** FACT
call_id, category (enum), quote, confidence

**`call_competitors`** FACT
call_id, competitor_name, context, sentiment

**`call_action_items`** FACT
call_id, owner_team, text, status

**`dim_account / contact / opportunity / rep`** DIM
mirrored from Salesforce IDs

**`workflow_run`** OPS
run_id, call_id, status, failed_node, ai_model, token_cost

**`ai_quality`** OPS
run_id, reviewer, thumb, notes — the eval loop

**`processed_calls`** OPS
fingerprint (unique), call_id, source — the atomic dedup claim

### Q Every Q2 call where pricing was the main objection

```
SELECT c.call_id, a.name, o.quote
FROM fact_call c
JOIN call_objections o USING(call_id)
JOIN dim_account a ON a.id=c.account_id
WHERE o.category='pricing'
  AND c.call_date BETWEEN '2026-04-01' AND '2026-06-30';
```

### Q Is Competitor X trending, month over month?

```
SELECT DATE_TRUNC(c.call_date,MONTH) m,
       COUNT(*) mentions
FROM call_competitors cc
JOIN fact_call c USING(call_id)
WHERE cc.competitor_name='Competitor X'
GROUP BY m ORDER BY m;
```

`Write path → the BigQuery Insert node writes fact_call + children; the MERGE node maintains processed_calls; every run (success or via the Error Trigger) writes workflow_run; the eval sampler writes ai_quality.`

### 🔒 Governance, PII & retention

Transcripts are PII and exec calls are restricted, so the store is governed like one: a **raw_transcript** dataset (admin-only, mirroring the Gong restricted area) kept apart from an **insights** dataset analysts query; row-level security by owner + column policy tags masking emails/names; CMEK encryption with region pinned for residency; time-partitioned **retention** (raw transcripts purged after N quarters) with GDPR erasure by `call_id`; and `workflow_run` as the audit/lineage trail. For a security company this is a day-one requirement, not an afterthought.

Gambit · GTM Engineer Take-Home — designed by **Omri Gonen**, 2026.

Built on n8n · Claude · Salesforce · Gong · Grain · BigQuery · Slack

Live submission · omriki55.github.io/jobmate-ai